Winter Wheat Volatility in the wake of the Russia-Ukraine War

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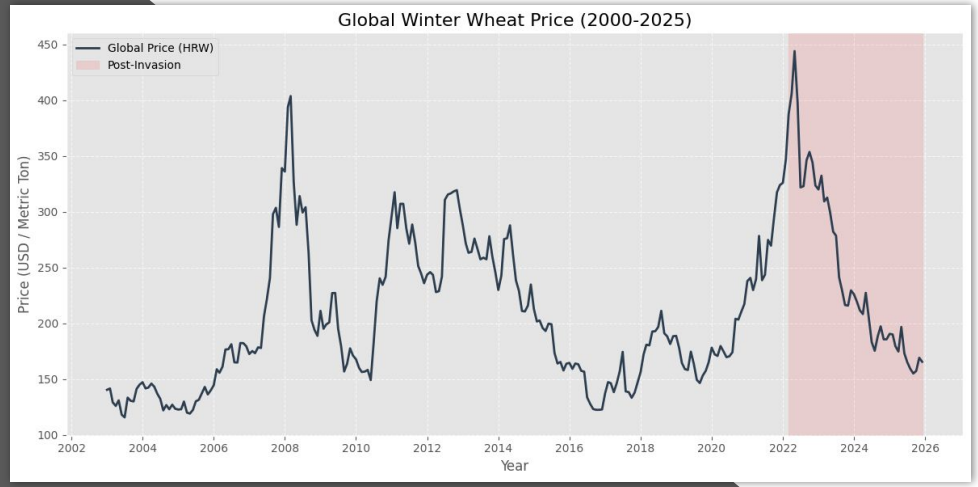
How does the Russia Ukraine war impact and continue to impact the world's wheat trade?



Price chart derived from data from:
International Monetary Fund, Global price of Wheat ,
retrieved from FRED (Federal Reserve Bank of St. Louis)
https://fred.stlouisfed.org/series/PWHEAMTUSDM

On the 24th of February 2022, Russia would invade Ukraine in what it called a "**special military operation**". The aftermath of which would send the global commodities market into a panic.

This analysis looks to examine the major economic effects of this event and the supply shocks felt throughout the global food system. Beyond the simple volatility in price, the study will attempt to map the physical displacement of trade to identify which nations stepped in to fill the export void, which import dependent regions bore the brunt of the shortage and whether strategic stockpiling behaviours leading up to the war exacerbated the situation.

Research Questions

The analysis aims to approach the data from several different perspective to uncover the impacts of war in one of the major producers of wheat in the world. To gain a clearer picture of the impact to global trade, we will consider questions such as:

- How has the invasion of Ukraine affected the global wheat prices?
- What was the strategic importance of the timing of the War?
- Has the price normalized again?
- Who gained the most from the War and disrupted supply of wheat?
- Has Ukraine's wheat industry recovered?
- What was the effect like for those without robust production capacity?

About the Data

The datasets used in this analysis come from two different sources. The first is the United States Department of Agriculture, specifically their Foreign Agricultural Service. This data is in the public domain and is shared under the **CC0 (Creative Commons Zero) License**.

The second dataset is price data for winter wheat. This was obtained from the International Monetary Fund, retrieved through FRED (Federal Reserve Bank of St. Louis). This data is licenced under **Fair use**, and allows for use in reporting and research purposes and allows free non-systematic downloading and/or printing of content from its sites.

The sources above are updated on a regular basis and are used at a federal level within organisations in the United States. A note must be made regarding limitations on Ending Stocks. These are calculated using residual data (Supply minus Demand) rather than physical count of grain in silos.

While price data is reported monthly, production data has a time lag and i reported on a yearly basis. As such, effort has been only make calculations on completed Market Years

Methodology

This analysis employs a qualitative approach to assess the impact of the 2022 invasion of Ukraine on the global wheat market. Annual country level metrics from the USDA Foreign Agricultural Service included Production, Yield, Area Harvested, Import, Exports and Ending Stocks. Monthly whea prices were gained from the Federal Reserve Bank of St. Louis. The specific benchmark used to represent price was US Hard Red Winter

Data Engineering and Exploration was done in Pandas and Matplotlib before the final visualizations were done in Datawrapper, which was chosen for its ability to render geospatial data and ease of use. Once clean, data was stored in Parque format to preserve data types and floating point precision which are often lost using CSV format.

Analysis and Findings

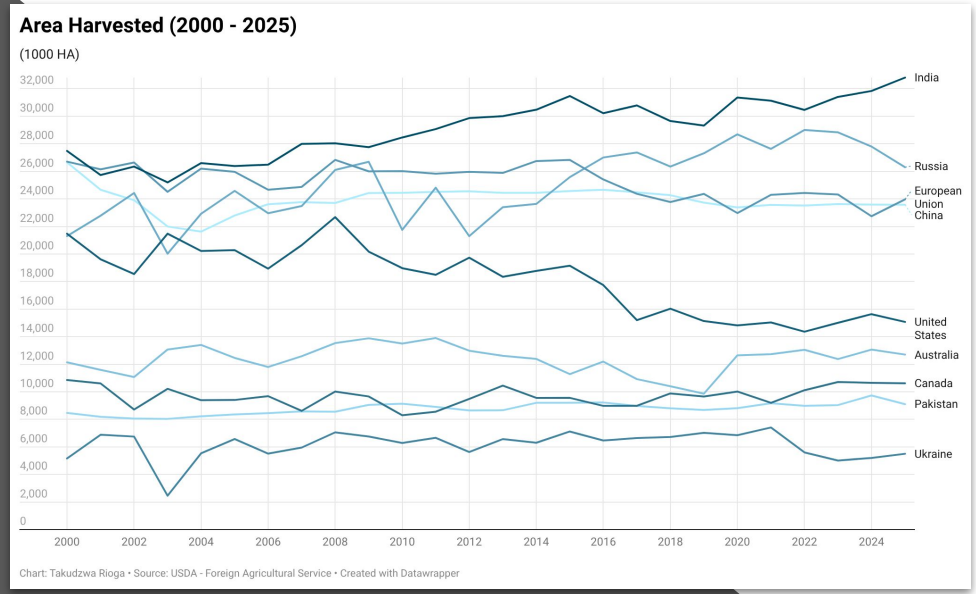
Ukraine's Wheat Production and Efficiency

Ukraine's Rise as a Global Player

Production Graph derived from data from:
USDA Foreign Agricultural Service

Ukraine has not always been a staple on the world stage for wheat and grains export but has had to develop its industry over the last 25 years. Ranging from the year 2000, production on average has increased in the region of about +793 tons per year before reaching a historic peak in 2021 with a recorded production of **33 Million Metric Tonnes**. Post 2022, this throughput has seen a marked decline and has been consistently hampered since.

Ukraine's Rise as a Global Player



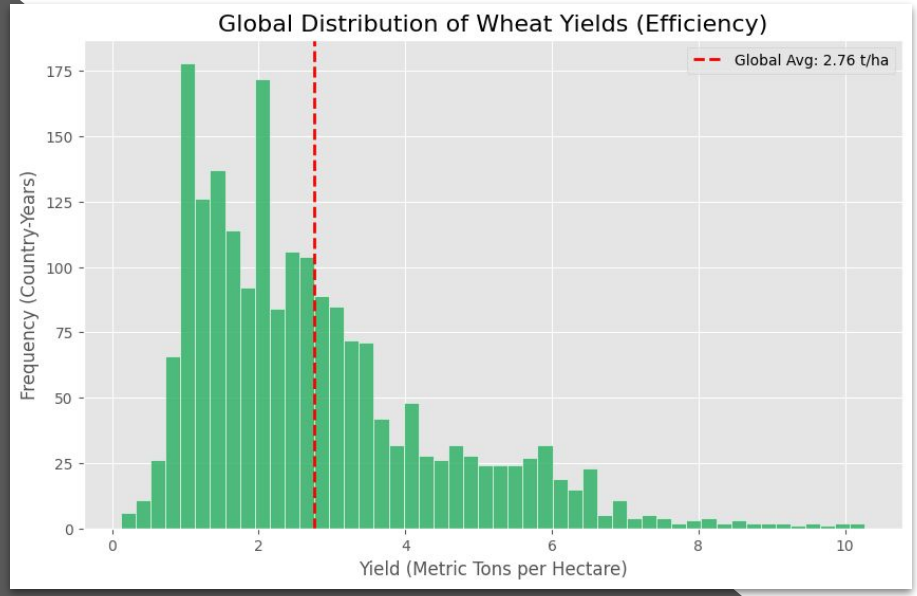
Land Use Trends derived from data from:
USDA Foreign Agricultural Service

In this graph, we see how Ukraine's harvested area has remained relatively consistent, pointing to advancements in technology and farming methods being the driving force behind it's increase in production and not a result in land reallocation.

A similar case can be made for the United States which continues to produce a large amount of the worlds wheat supply even though they have scaled back land allocation for wheat farming.

Conversely India has seen the largest increase in land reserved for growing along with Russia, which between 2012 and the start of the conflict with Ukraine has steadily increased its allocation.

Ukraine's Rise as a Global Player

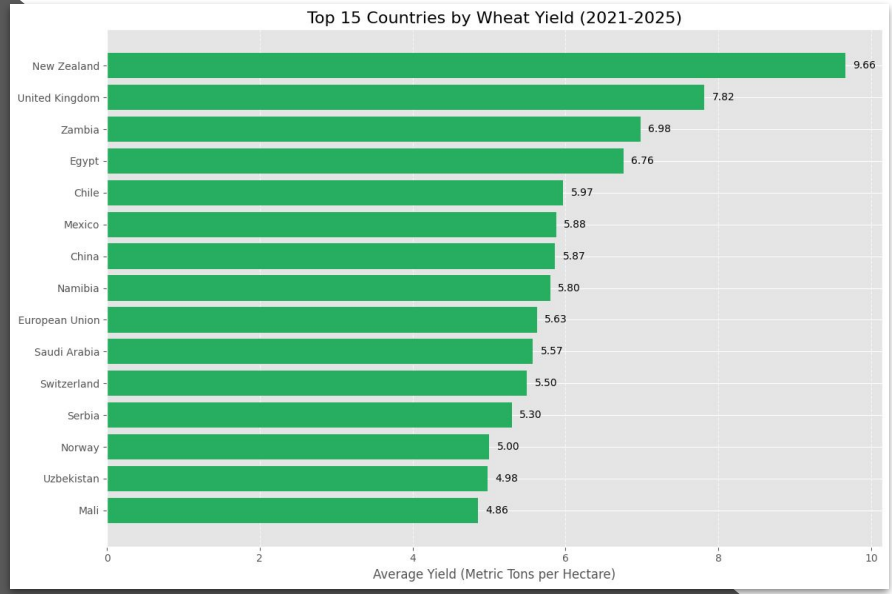


Efficiency Graph derived from data from:
USDA Foreign Agricultural Service

To better understand the magnitude of this increase in productivity, we can turn to the global efficiency of other countries. Efficiency here is measured as the amount of wheat produced in tonnes per hectare of land.

From this, on average, we can expect most countries to produce close to 3 tonnes of wheat per hectare of land with clear outliers producing close to 10 tons per hectare.

Ukraine's Rise as a Global Player

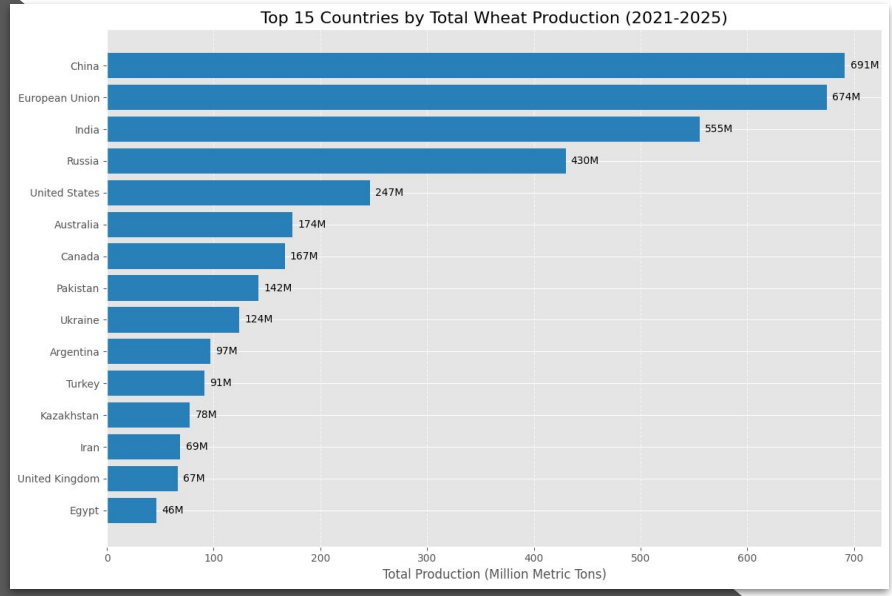


Efficiency Chart derived from data from:
USDA Foreign Agricultural Service

We see with the exception of China and the European Union (which is made up of a collection of different countries), the United Kingdom and Egypt, no other country can be considered a highly efficient producer **as well as** a major producer capable of generating significant export volumes.

The specialization required to master both concentration and volume is a unique skill.

Ukraine's Rise as a Global Player

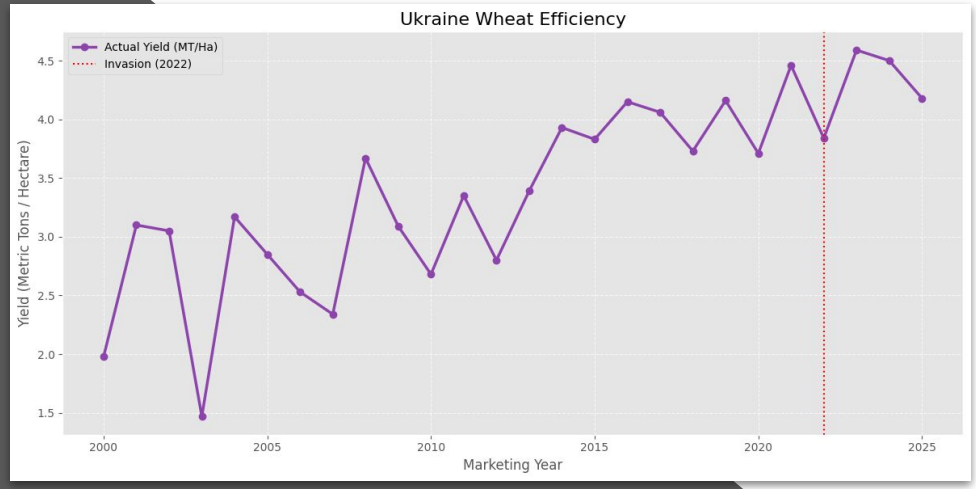


5 Year Production Graph derived from data from:
USDA Foreign Agricultural Service

This graph further illustrates the point by listing the top producers in the last 5 years. As we see clearly, only those four countries mentioned before also appear on this list. Namely:

- China
- The European Union
- The United Kingdom
- Egypt

Ukraine's Rise as a Global Player



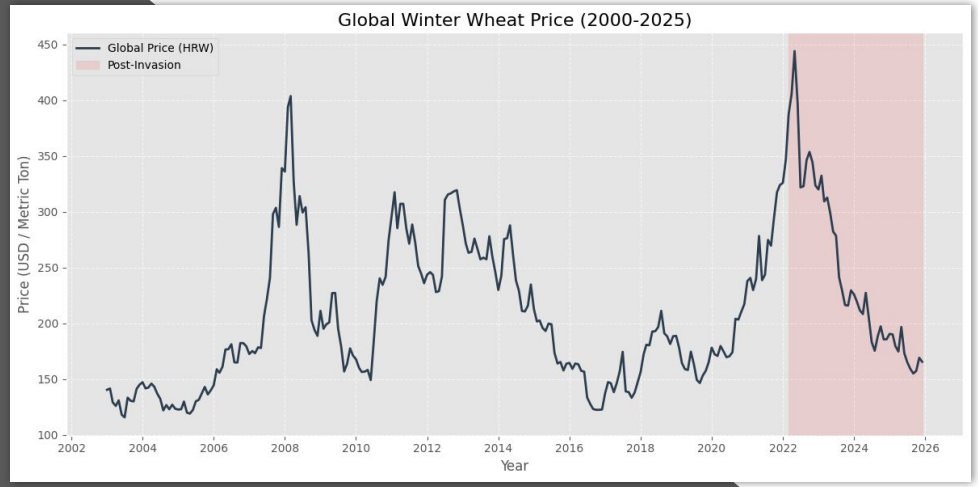
Ukraine Efficiency Graph derived from data from:
USDA Foreign Agricultural Service

Though it should be noted that Ukraine has made significant strides towards becoming both a highly efficient producer as well as a one capable of exporting large quantities of wheat to the global market.

In this graph, we see how Ukraine's steady increase in efficiency over time.

An Over Reaction From the Market...?

Supply Chain Disruption



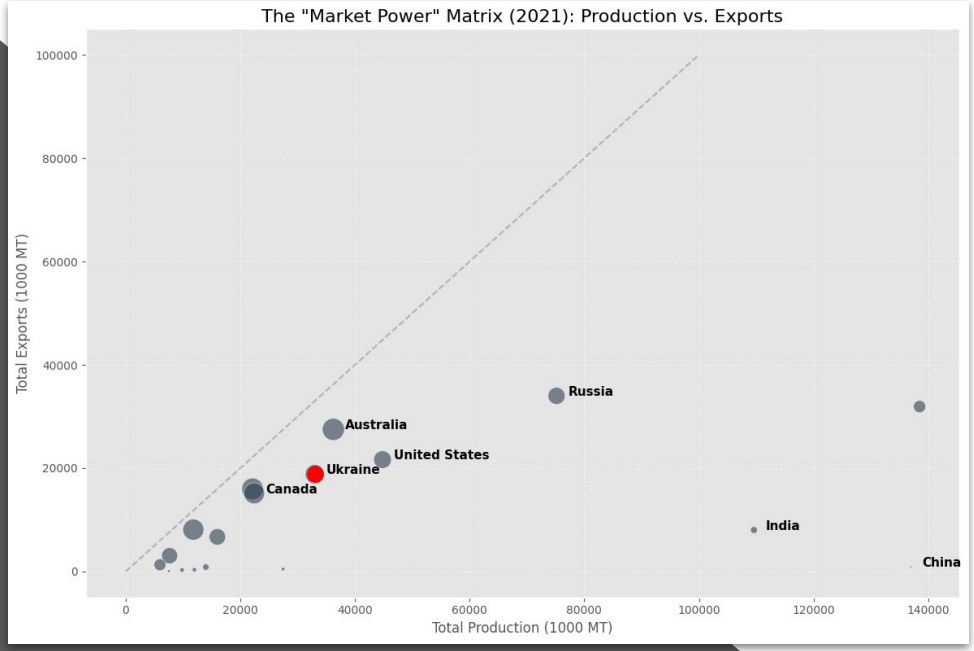
Price Chart derived from data from:
USDA Foreign Agricultural Service

As we've just seen, Ukraine does not produce the most wheat. But to understand the markets reaction, we need to understand what makes a country visible in the global commodities market.

China undoubtedly has the largest production of any single country by some margin, though Ukraine is the one moving the market and causing supply chain disruptions and bring about the highest historical price for wheat in recorded history?

To answer this we must understand "**Market Power**".

Supply Chain Disruption

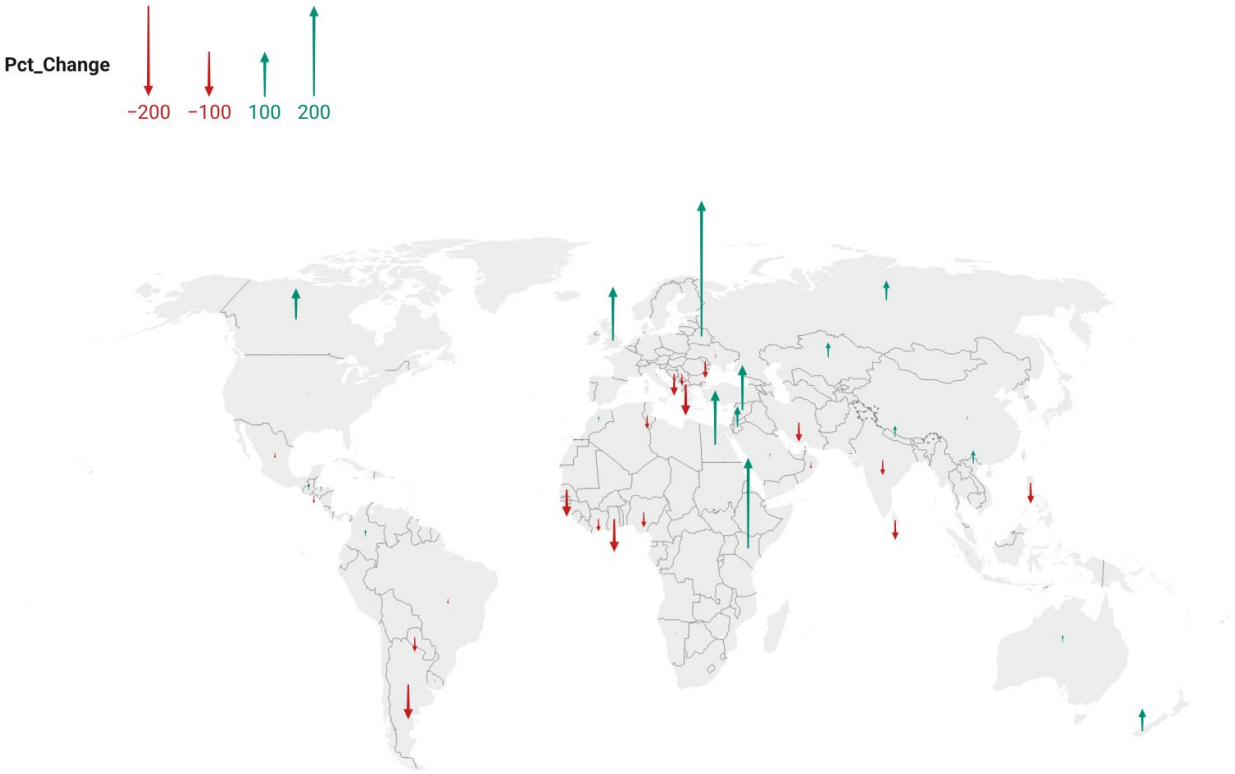


Export to Production Graph derived from data from:
USDA Foreign Agricultural Service

Unlike Ukraine, China is entirely focussed on feeding its own population and satisfying its own domestic market. We see that on a Production to Export basis, Ukraine is heavy focussed on moving as much of it's wheat into the market.

The closer a country is to the 45 degree slope, the greater its share of wheat production does not stay within its borders. This is the greatest reason why Ukraine on paper does not produce as much but creates such waves in the market.

Percentage Change of World Wheat Export (2021 to 2022)

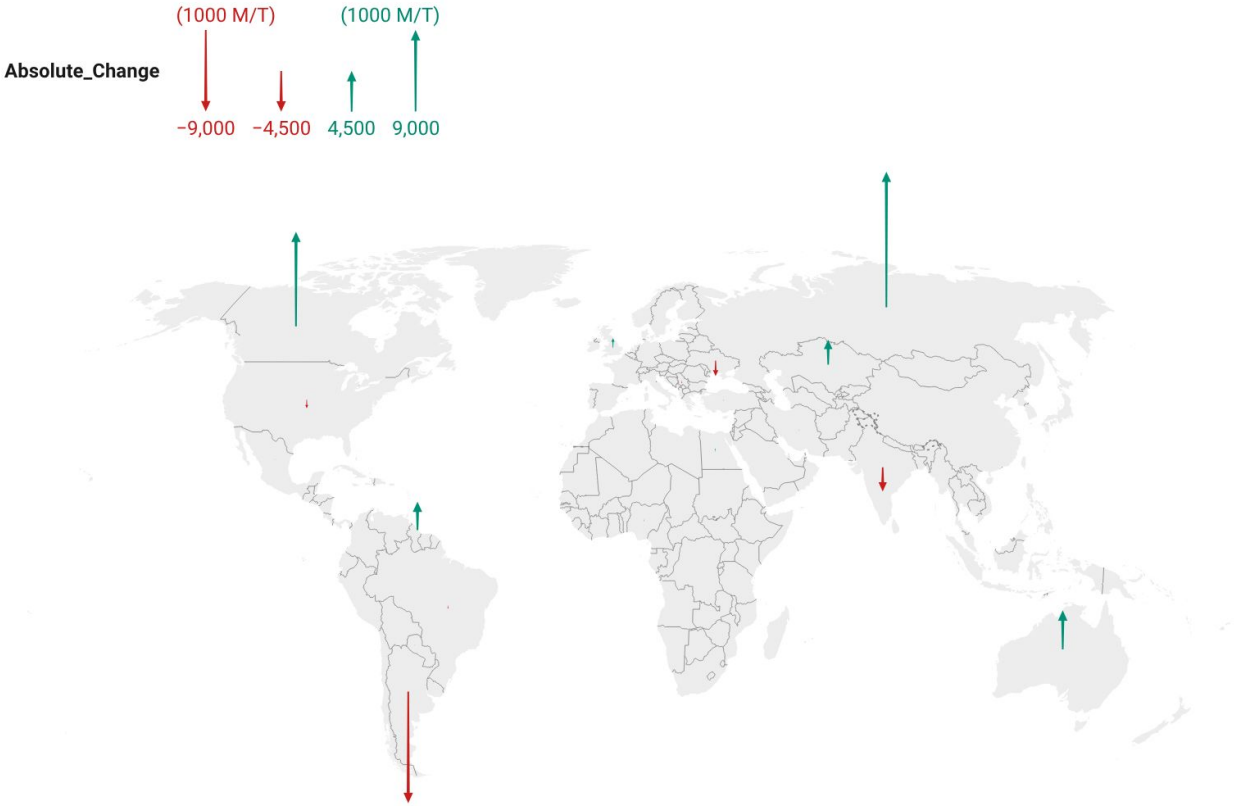


As we see, different economies are forced to react differently depending on their ability to either produce wheat of their own or their need to restructure their supply chains to source wheat elsewhere.

The greater the percentage change, the great a nation has had to pivot from it old Import/Export strategies.

Map: Takudzwa Rioga • Source: USDA - Foreign Agricultural Service • Created with Datawrapper

Absolute Change of World Wheat Export (2021 to 2022)



Here we see in absolute terms which nations have succeeded in either filling the gap left in the market by the absence of Ukraine Wheat or have had to close off their commodity export channels.

Countries like Australia, Russia, Canada can now benefit from the increased demand for their grains.

Argentina is the anomaly in this case. Their retreat from the global export market is driven primarily due to drought experienced in the early 2020's. Though undoubtedly the disruptions to supply added their own measure of uncertainty.

Map: Takudzwa Rioga • Source: USDA - Foreign Agricultural Service • Created with Datawrapper

Change in Global Import Volumes

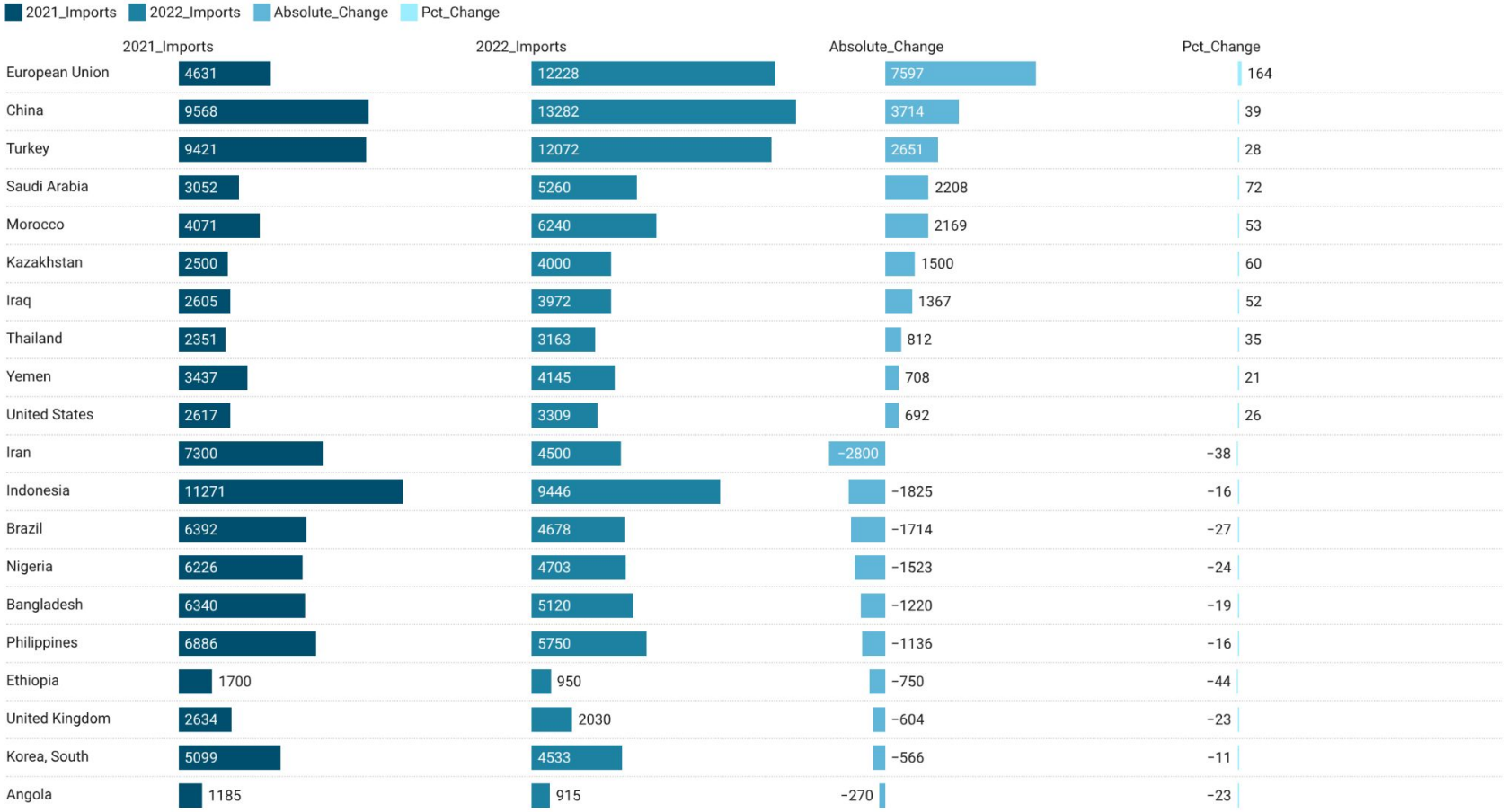


Chart: Takudzwa Rioga • Source: USDA - Foreign Agricultural Service • Created with Datawrapper

A Preemptive Move by Russia?

Export to Production Graph derived from data from:
USDA Foreign Agricultural Service

The one question we have yet to answer is whether Russia deliberately held onto an above average level of wheat in preemption of their invasion.

The data would suggest not. Though Russia did see a large increase in its wheat export by absolute value, the production to export ratio does not suggest anything to that effect. Retained stocks have been trending constantly since the year 2000.

There deviation from the 3 year mean (**on the next page**) does suggest some move away from the mean by means of a 37 percent above average Ending Stock, we cannot draw clear conclusions.

Deviation from 3 Year Ending Stock Baseline

	Baseline_Avg	2021_Stocks	Excess_Stocks	Pct_Increase
Ukraine	1521	6265	4744	312
Russia	8795	12088	3293	37
Pakistan	2193	4530	2337	107
European Union	13202	13631	429	3
Iran	4686	5086	400	9
Egypt	4971	5330	359	7
Ethiopia	913	1153	240	26
Kazakhstan	1284	1479	195	15
Indonesia	1590	1700	110	7
Japan	1075	1176	101	9

Chart: Takudzwa Rioga • Source: USDA - Foreign Agricultural Service • Created with Datawrapper

LICENCE

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